

# BCOG 100

Mind and Brain Lecture 2:  
Levels of Analysis



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# Starting question

**Password:** three

**Question:** What are David Marr's Three Levels of analysis.  
Give a definition of each in your own words.



# Mind-Body Solutions

## Two Dimensions of Disagreement

1. **Ontological Dependence:** Is the mind a different kind of “stuff” than the physical, material world?
2. **Explanatory Separability:** Can we understand the physical and the mental independently (whatever they are made of)?

|                            |     | Ontologically Dependent? |    |
|----------------------------|-----|--------------------------|----|
|                            |     | Yes                      | No |
| Explanatorily<br>Separable | Yes | 4                        | 1  |
|                            | No  | 2                        | 3  |

# Mind-Body Solutions: Quadrant 4

## The Popularity of Quadrant 4:

- Still materialist and scientific (as opposed to quadrant 1 and 3)
- Psychological explanations work
- Theoretical limitations: can you really eliminate all talk of memory, attention, concepts?
- Most sciences are reductive without being eliminativist
- Practical constraints. modeling in terms of neurons is intractable
- Multiple realizability



# Mind-Body Solutions: Quadrant 4

Ontologically Dependent?

|                            |     | Yes | No |
|----------------------------|-----|-----|----|
| Explanatorily<br>Separable | Yes | 4   |    |
|                            | No  |     |    |

**Definition:** Mind depends on physical systems, but can sometimes be explained in terms that are not reduced to physical ones.

## Example: Functionalism

- What makes something a mental state is what it does, not what it's made of. Pain = the state that's caused by injury and leads to avoidance, regardless of whether it's in a brain, a robot, or an alien.
- Mental states are defined by their functions or roles in a system, not their physical substance.

**Problem:** It struggles with qualia — explaining why these functional roles are accompanied by felt experience (“what it’s like” to be in pain).

# Mind-Body Solutions: Quadrant 4

Ontologically Dependent?

|                            |     | Yes | No |
|----------------------------|-----|-----|----|
| Explanatorily<br>Separable | Yes | 4   |    |
|                            | No  |     |    |

**Definition:** Mind depends on physical systems, but can sometimes be explained in terms that are not reduced to physical ones.

## Example: Emergentism

- Mental phenomena emerge from complex physical systems like the brain, the way liquidity emerges from H<sub>2</sub>O molecules.
- The mind is not separate from the brain, but it has new properties that aren't reducible to the properties of neurons.

**Problem:** Critics ask whether “emergence” really explains anything — or if it’s just a label for our ignorance about how brain activity produces cognition.

# Mind-Body Solutions: Quadrant 4

Ontologically Dependent?

|                            |     | Yes | No |
|----------------------------|-----|-----|----|
| Explanatorily<br>Separable | Yes | 4   |    |
|                            | No  |     |    |

**Definition:** Mind depends on physical systems, but can sometimes be explained in terms that are not reduced to physical ones.

## Example: Multiple Levels of Analysis

- To understand the mind, we must explain it at different levels:
- Each level gives a valid, necessary kind of explanation; no single level captures the whole picture.

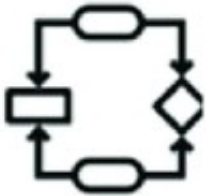
**Problem:** How do we connect the levels? For example, how do abstract computations (like vision) map onto specific neural processes?



# David Marr's Levels of Analysis



**Computational level:** What is the system doing? What is the problem being solved?



**Algorithmic level:** How is it being done? What representations and processes are used?



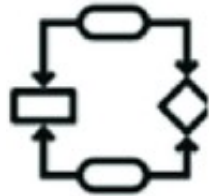
**Implementational level:** Where and by what physical means is it carried out?



# Example: Vision



**Computational level:** The goal is to recover 3D structure of the world from 2D retinal images.

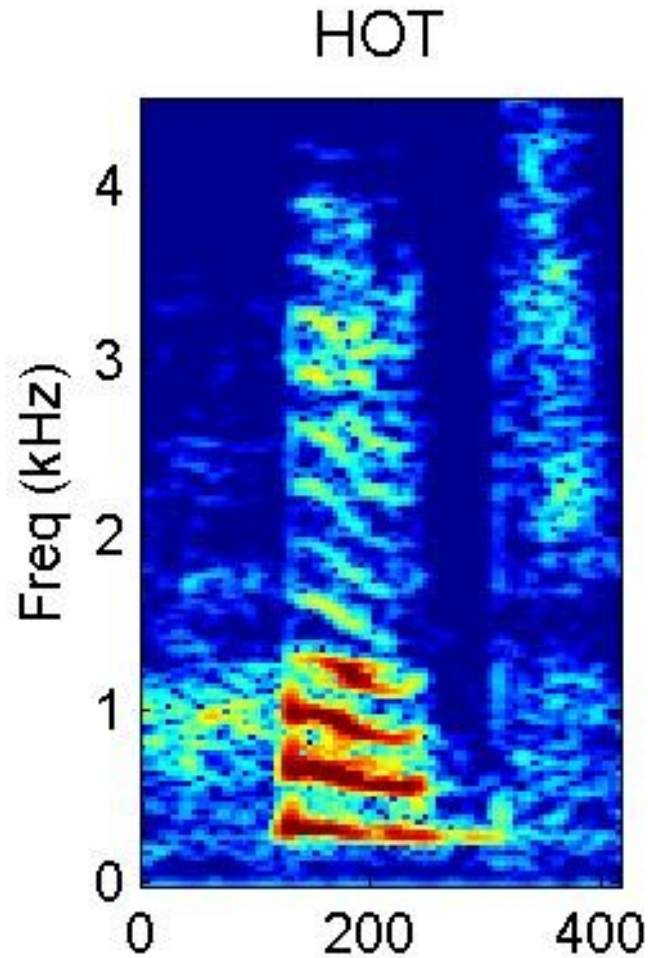


**Algorithmic level:** Edge detection, depth cues, object segmentation, recognition processes.

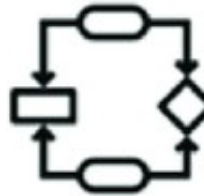


**Implementational level:** Retinal cells, LGN pathways, cortical neurons in V1, V2, IT, etc.

# Example: Language



**Computational level:** Map an acoustic signal onto a speaker's intended message.



**Algorithmic level:** Segment words, determine syntactic structure, compute meaning of whole utterance.



**Implementational level:** Neurons in auditory cortex, temporal lobe, Broca's area, recurrent neural networks, etc.

# Strengths and Benefits of Marr's Approach

- Encourages **clarity**: distinguish *what* from *how* from *where*.
- Supports **explanatory separability**: cognitive models don't have to map directly to neurons.
- Helps avoid **reductionist traps** (e.g., dismissing psychology because we have neuroscience).
- Provides a **division of labor** across disciplines (psychology, neuroscience, AI, philosophy).



# Challenges to Marr's Approach

- Rigid separation: In practice, levels are not always independent.
- Top-down bias: Assumes the computational problem is well-defined in advance.
- Real systems are messy: Evolution doesn't optimize cleanly for abstract "functions."
- Example: modern AI shows implementational details (architecture, data statistics) shape what algorithms emerge.



# Challenges to Marr's Approach

## **Marr's view:**

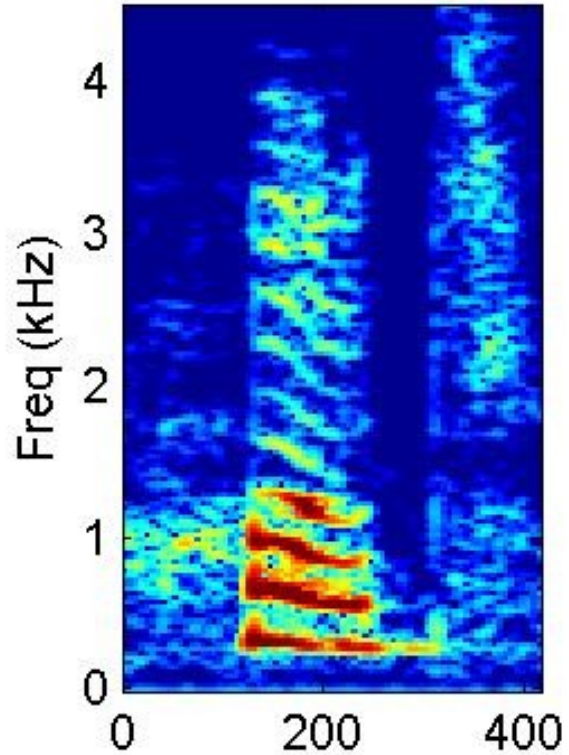
- Computational: The goal is to store experiences and retrieve them later.
- Algorithmic: Processes like encoding, storage, and retrieval.
- Implementational: Brain regions like the hippocampus and cortex.

## **Problem:**

- The way memory works depends heavily on the brain's wiring.
- For example: The hippocampus is especially good at quickly linking events together, but struggles with long-term storage on its own.
- This means the “algorithm” is shaped by the biology — you can't neatly separate the levels.

# Another Way of Thinking About Levels

HOT



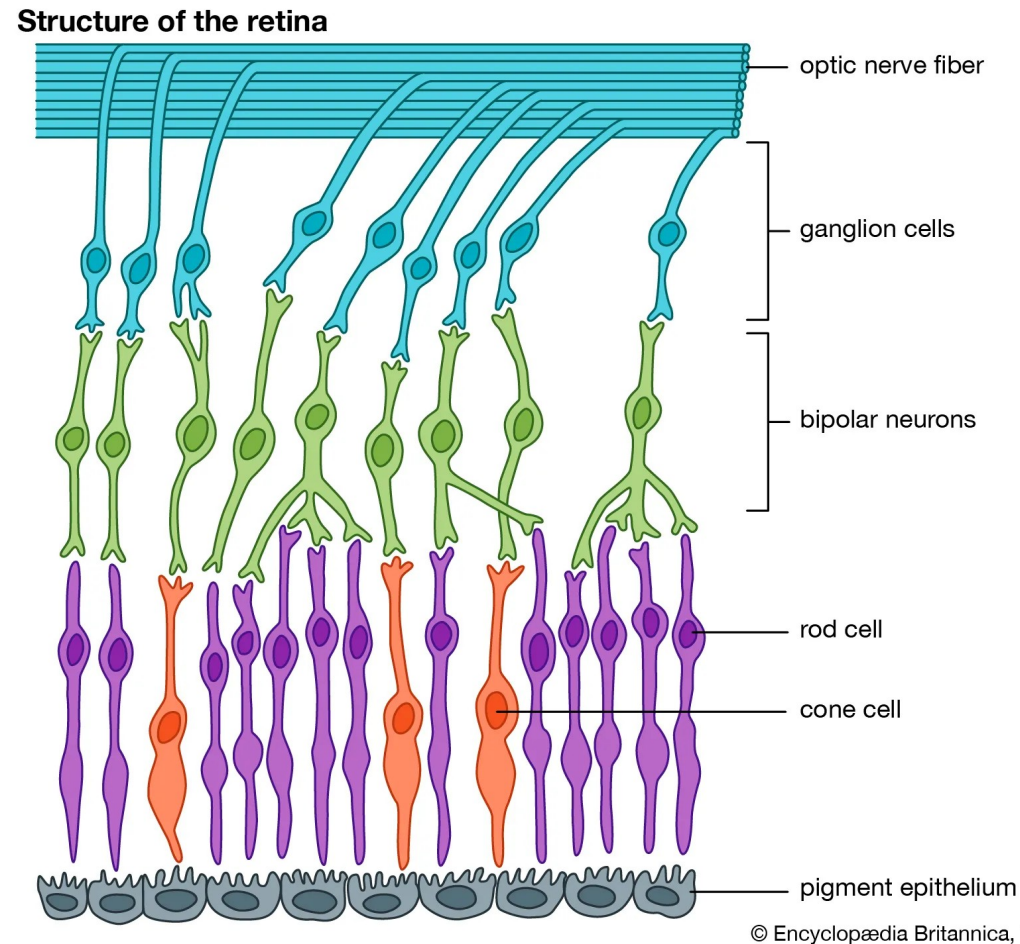
There are a lot of different ways you can move your mouth and say “hot”

There are a lot of different auditory vibrations you can detect and hear “hot”

Is speaking or hearing “hot” a “real” thing?

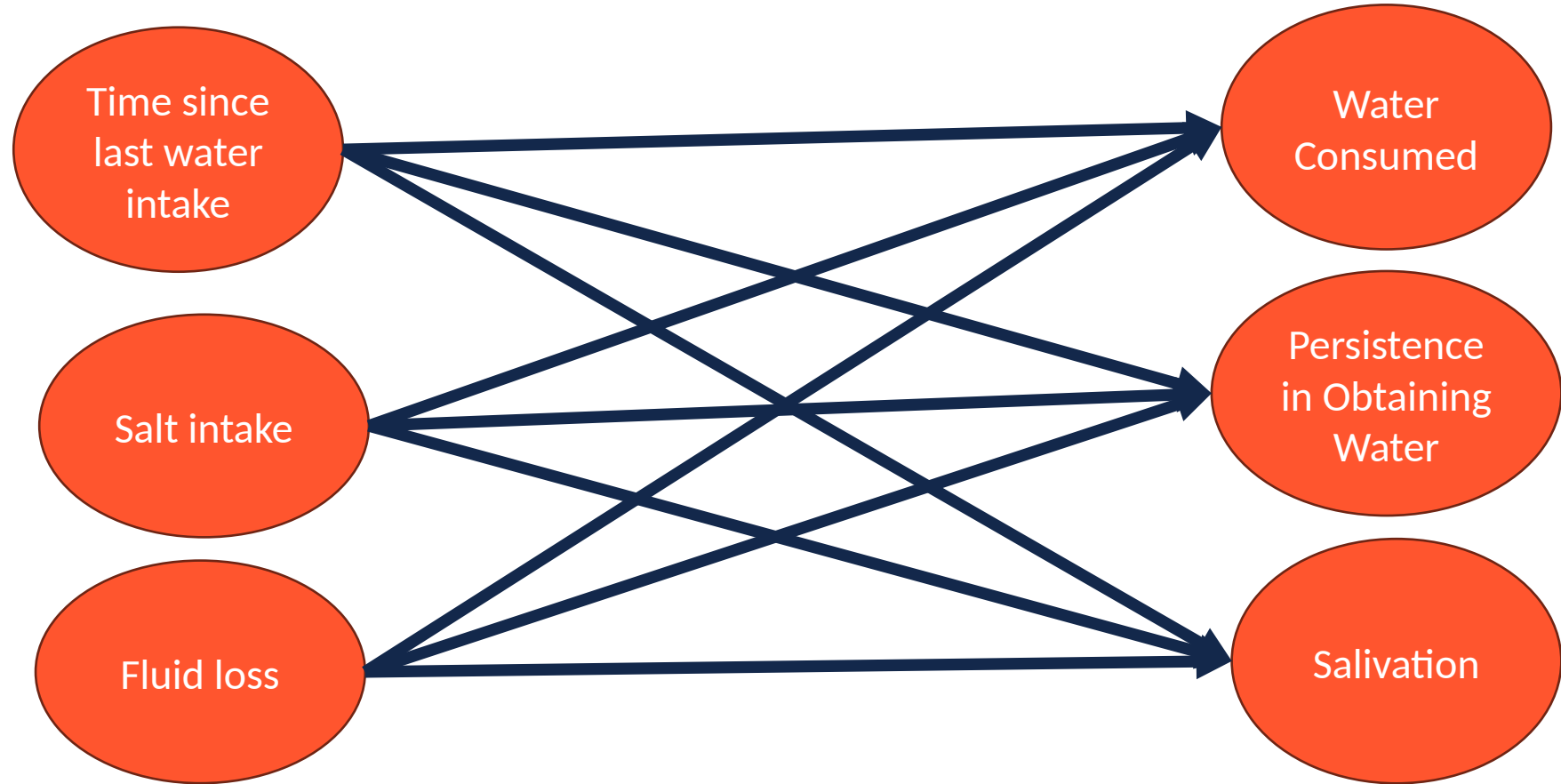


# Another Way of Thinking About Levels



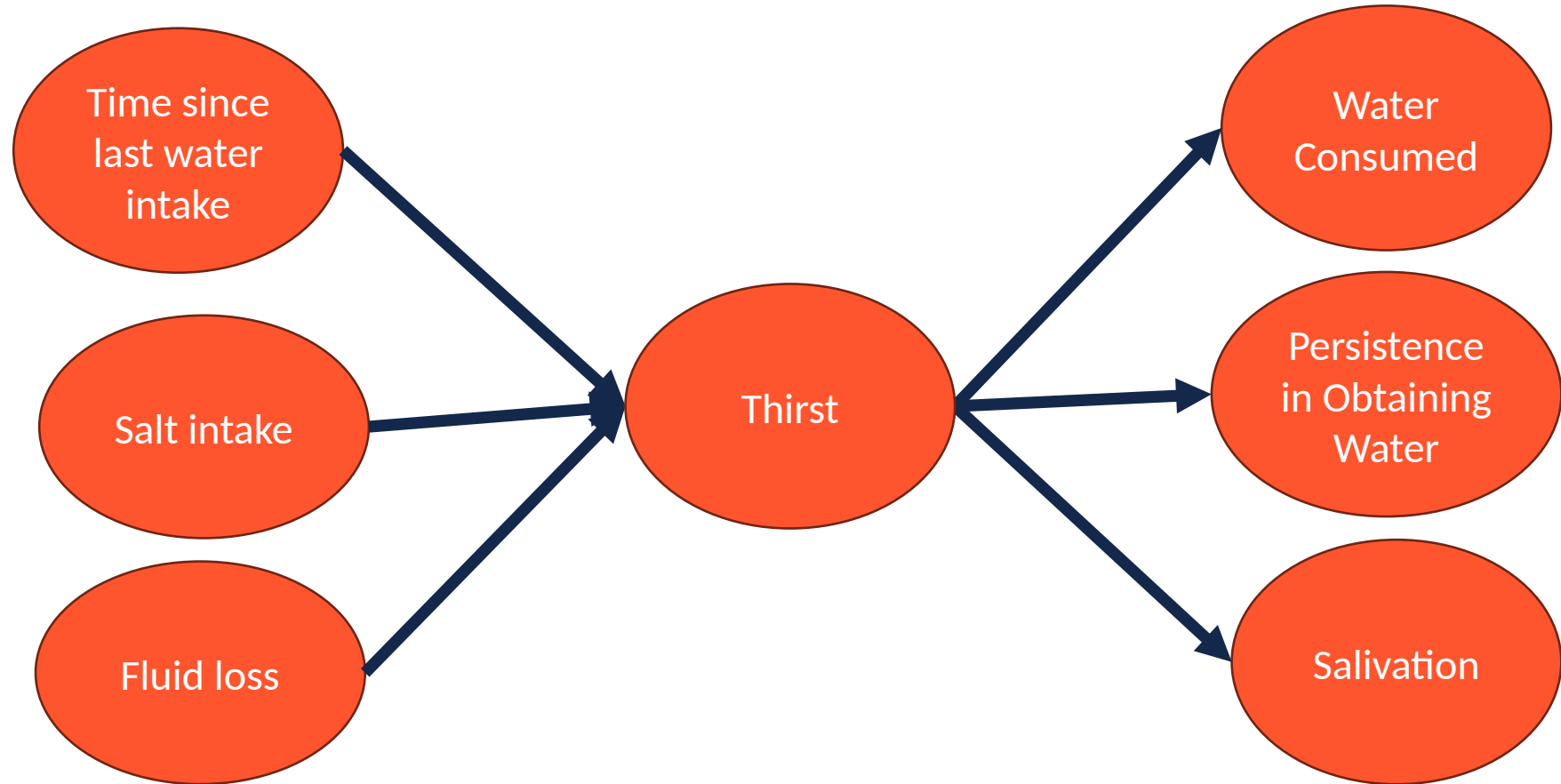
Is the concept useful? Does it lead to predicting things as well, with less complexity?

# Another Way of Thinking About Levels



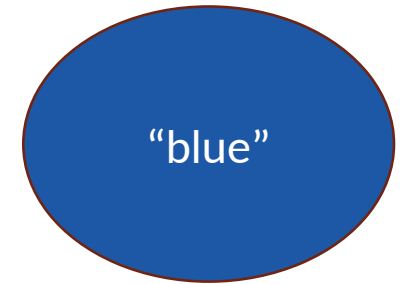
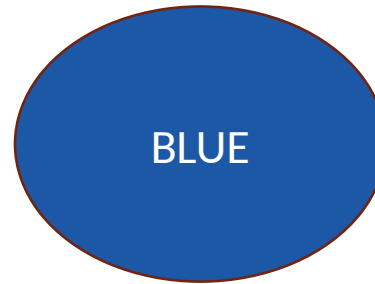
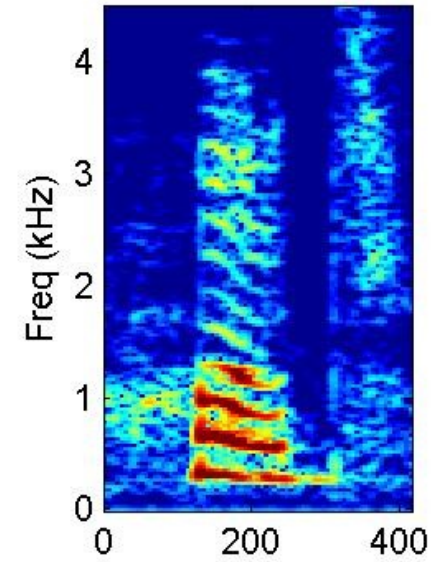
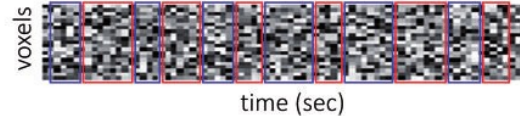
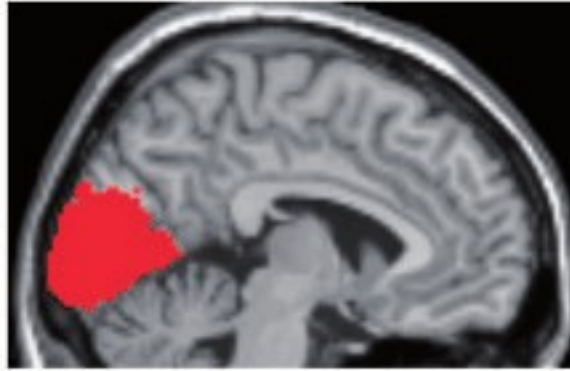
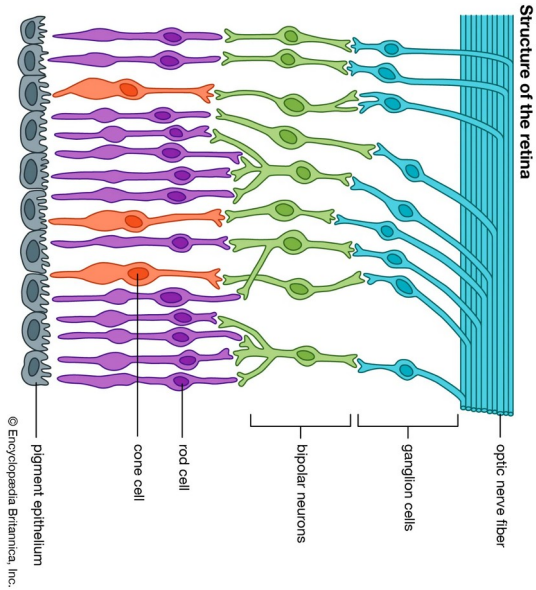


# Another Way of Thinking About Levels



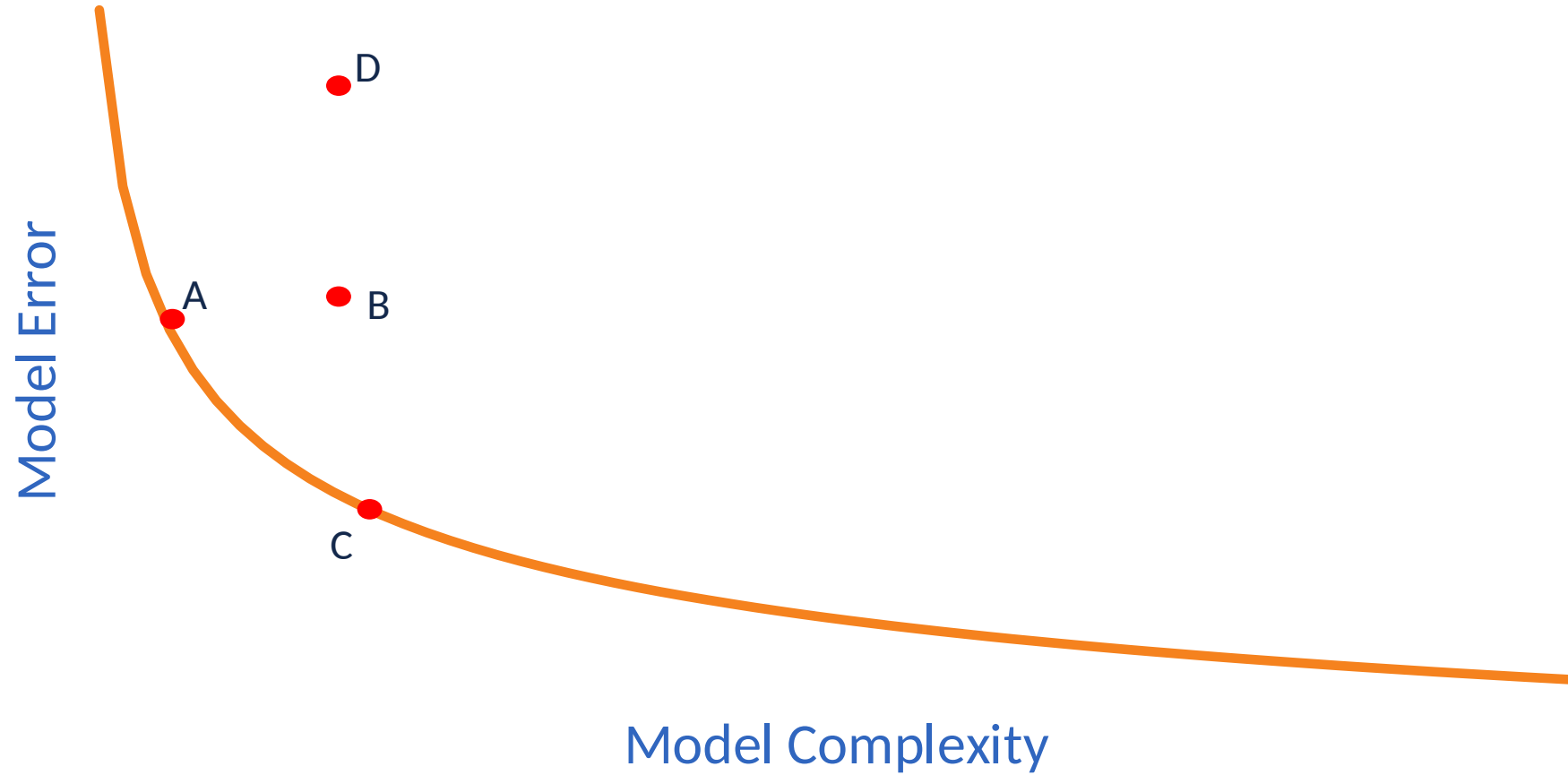
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# Another Way of Thinking About Levels



Is the concept useful? Does it lead to predicting things as well, with less complexity?

# Another Way of Thinking About Levels



# Summary

- Marr's levels remain hugely influential.
- Most cognitive sciences endorse multi-level explanations.
- But modern work highlights interactions between levels: not cleanly separable, often mutually constraining.
- Most scientists are not dogmatic about what the levels are. The focus is more pragmatic.
  - Do your concepts make good predictions
  - Do they help lead to understanding

